

Experiment on 8051 Microcontroller using EdSim51

Experiment 1

Question: Write a program to add two numbers 05H and 07H.

The screenshot displays the EdSim51 software interface, which is divided into several sections:

- System Configuration:** Located at the top left, it includes settings for System Clock (MHz), SBUF, and various registers (R0-R7, B, ACC, PSW, IP, IE, PCON, DPH, DPL, SP, PC).
- Program Editor:** The central area shows the assembly code being executed. The code is as follows:

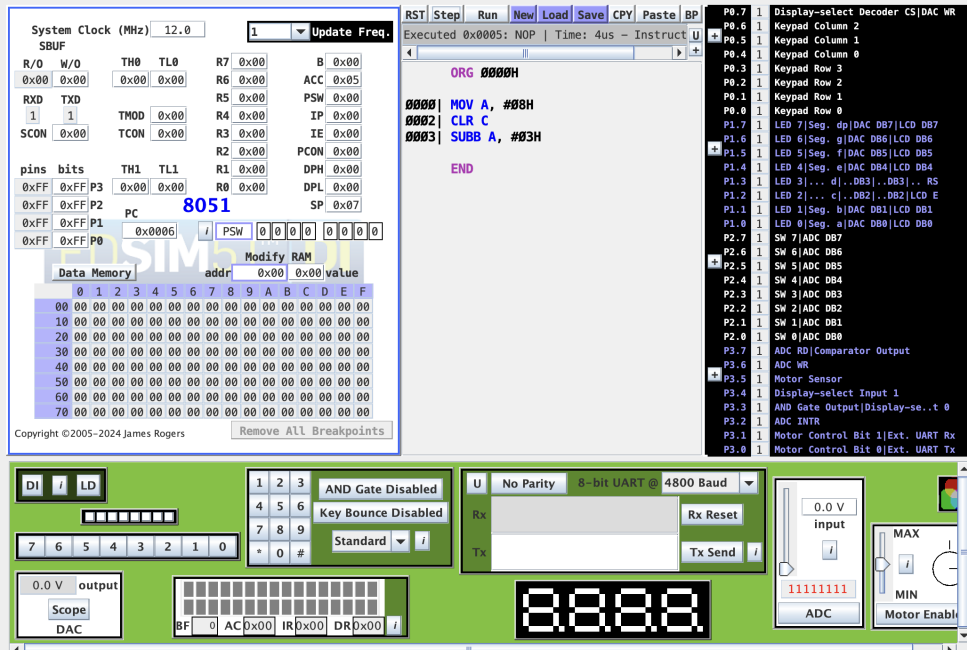
```
ORG 0000H
0000 MOV A, #05H
0002 MOV B, #07H
0005 ADD A, B
0007 MOV 30H, A
END
```
- Execution/Output:** The bottom section shows the results of the program execution. It includes a digital display showing '8.8.8.8', a keypad, and various control buttons like 'AND Gate Disabled', 'Key Bounce Disabled', 'Standard', 'Rx Reset', 'Tx Send', 'ADC', and 'Motor Enabled'. The 'ADC' section shows a value of '11111111'.

EdSim51 program and execution/output screenshot.

Experiment on 8051 Microcontroller using EdSim51

Experiment 2

Question: Write a program for subtraction of two numbers.



EdSim51 program and execution/output screenshot.

Experiment on 8051 Microcontroller using EdSim51

Experiment 3

Question: Write a program for multiplication of two numbers.

The screenshot displays the EdSim51 software interface, which is used for simulating 8051 microcontroller programs. The interface is divided into several sections:

- System Configuration:** Located at the top left, it includes a 'System Clock (MHz)' set to 12.0 and an 'Update Freq.' button. Below this are various registers and flags like R7, R6, R5, R4, R3, R2, R1, R0, B, ACC, PSW, IP, IE, PCON, DPH, DPL, and SP.
- Program Editor:** The central area shows the assembly code being executed. The code is as follows:

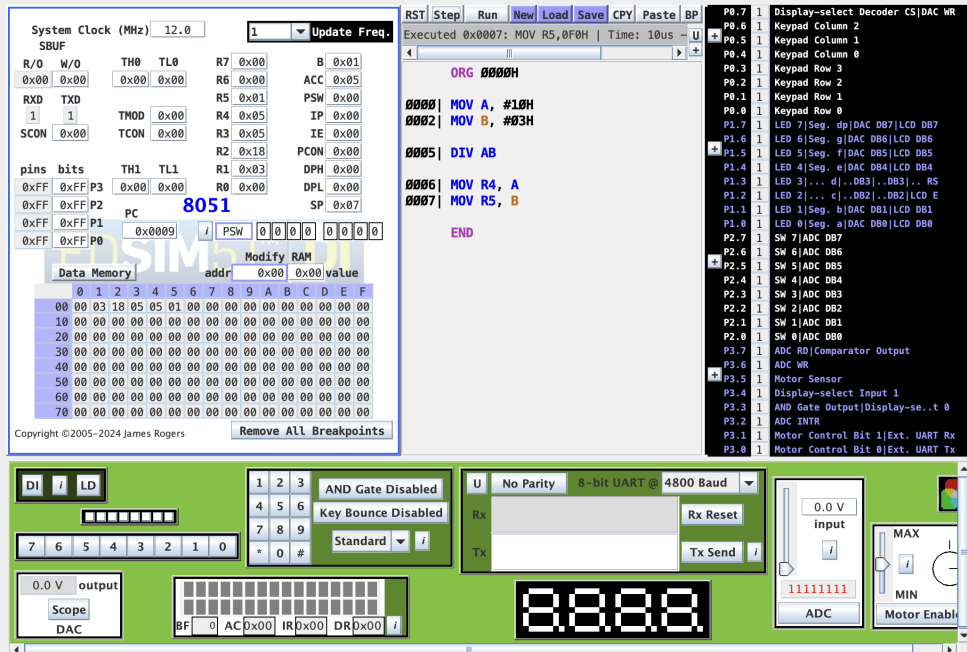
```
ORG 0000H
0000 MOV A, #05H
0002 MOV B, #07H
0005 MUL AB
0006 MOV 30H, A
END
```
- Hardware Simulation:** The bottom section shows a virtual hardware environment. It includes a keyboard (0-9, *, #), a numeric keypad, a display showing '11111111', an ADC (Analog-to-Digital Converter) with a value of 11111111, and a motor control section with a 'Motor Enabled' button.
- Registers and Memory:** On the right side, there is a list of registers and memory locations, including P0.7, P0.6, P0.5, P0.4, P0.3, P0.2, P0.1, P0.0, P1.7, P1.6, P1.5, P1.4, P1.3, P1.2, P1.1, P1.0, P2.7, P2.6, P2.5, P2.4, P2.3, P2.2, P2.1, P2.0, P3.7, P3.6, P3.5, P3.4, P3.3, P3.2, P3.1, and P3.0.

EdSim51 program and execution/output screenshot.

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Experiment 4

Question: Write a program to divide 10H with 03H and save the quotient in R4 and remainder in R5.

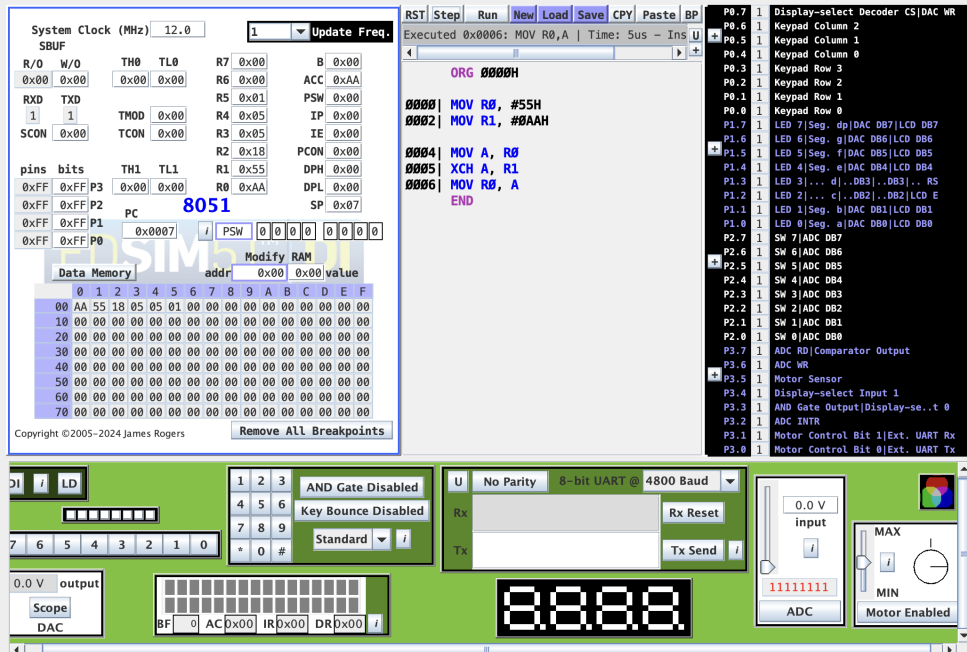


EdSim51 program and execution/output screenshot.

Experiment on 8051 Microcontroller using EdSim51

Experiment 5

Question: Write a program to swap two numbers: 55H is stored in R0 and AAH is stored in R1. Swap the contents of R0 and R1.



EdSim51 program and execution/output screenshot.

Experiment on 8051 Microcontroller using EdSim51

Experiment 6

Question: Write a program to blink the LED on Port 1.

The screenshot displays the EdSim51 software interface for an 8051 microcontroller. The top-left pane shows the system configuration, including a 12.0 MHz clock and various registers (R0-R7, B, ACC, PSW, IP, IE, PCON, DPH, DPL, SP). The top-right pane shows the assembly code for a program that blinks an LED on Port 1. The code starts with an ORG directive at 0000H, followed by a SETB P1.0 instruction. A BLINK loop is defined, consisting of a CPL P1.0 instruction, an ACALL DELAY instruction, and a SJMP BLINK instruction. The DELAY loop is implemented using a MOV R7, #0FFH instruction, followed by a loop of DJNZ R7, \$ and RET instructions. The bottom-left pane shows the data memory, and the bottom-right pane shows the execution/output, including a digital display showing 8.8.8.8 and a scope showing the output of the DAC.

```
ORG 0000H
SETB P1.0

BLINK:
CPL P1.0
ACALL DELAY
SJMP BLINK

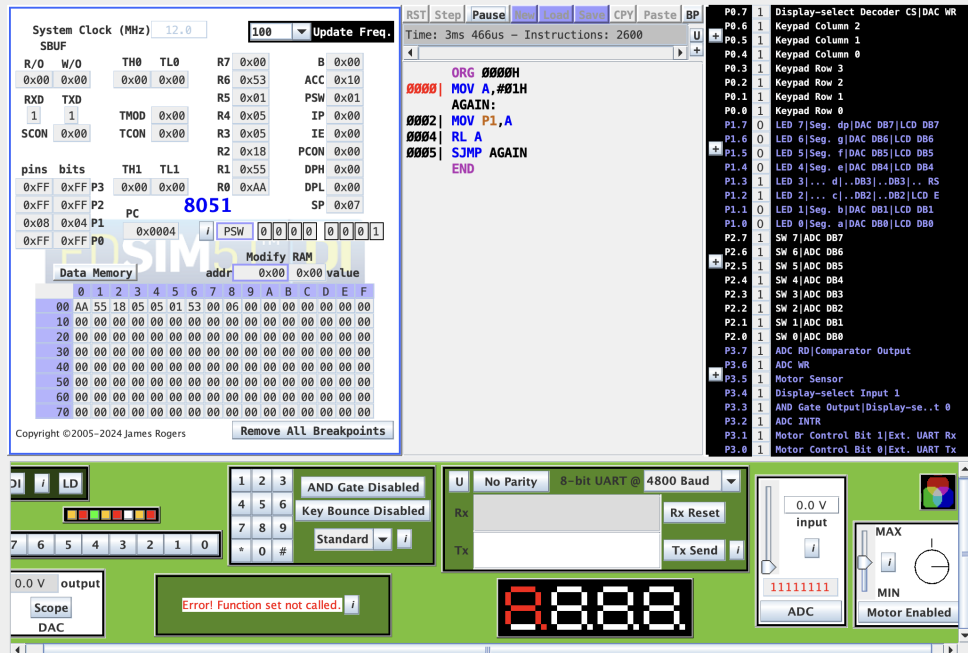
DELAY:
MOV R7, #0FFH
MOV R6, #0FFH
DJNZ R7, $
DJNZ R6, $
RET
END
```

EdSim51 program and execution/output screenshot.

Experiment on 8051 Microcontroller using EdSim51

Experiment 7

Question: Write a program to blink the LED from right to left.



EdSim51 program and execution/output screenshot.

